POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2020
Programme: B E | Full Marks: 100
Course: Physics | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Show that motion of a compound pendulum is angular simple harmonic and derive its time period. Also show that the time period of compound pendulum is minimum when length of pendulum is equal to the radius of gyration. 9

   b) A piano wire with mass 5gm and length 90 cm is stretched with tension of 25 N. A wave with frequency 100 Hz and amplitude 1.6 mm travels along the wire a) calculate the average power carried by the wave b) what happens to the power if the wave amplitude is halved. 6

2. a) What is interference of light wave? Describe the intensity distribution in the interference due to Young's double slits and show that the fringe width is same both for bright and dark fringes 9

   b) Derive the expression for the refractive index of the liquid which replaces the air film between Plano-convex lens and base plate in Newton's Ring experiment. 6

3. a) Define electric field and potential due to monopole, show that the magnitude of electric field along axial line due to linear dipole is double than that of the electric field due to dipole along equatorial line. 9

   b) Obtain the time constant of a capacitor in RC circuit such that the current through the resistor increases to one third of its peak value in 5 seconds. 6

4. a) State Biot-Savart's law. Use it to find the magnetic field due to an infinitely long straight wire. 9

   b) A circular loop of wire 5 cm in radius carries a current of 100 A. What is the energy density at the centre of the loop? 6

5. a) Write Maxwell's equations in integral and differential form. Using Maxwell's equation in free space show that speed of electromagnetic wave is equals to the speed of light. 9

   b) Discuss the LC oscillation. Hence prove that the frequency of oscillation is $f=\frac{1}{2\pi\sqrt{LC}}$. 6

6. a) Define wave function with significance. Prove that energy levels are quantized, when a particle is trapped in an infinite potential well of a width "a" such that. 9

   $V = \infty \; for \; x \leq 0 \; and \; x \geq a$

   $V = 0 \; for \; 0 < x < a.$

   b) The time of reverberation of an empty hall without and with 500 audiences is 1.5 sec and 1.4 sec respectively. Find the reverberation time with 800 audiences. 6

7. Write short notes on: **(Any two)** 2×5

   a) Atomic View of resistivity

   b) Types of optical fiber

   c) Semi-conductor dipping.